

Adaptive Control Applied to Static Converters with the Virtual Plant Method

Vinícius Descovi Rodrigues
*Núcleo de Pesq. e Desenvolvimento
em Engenharia Elétrica - NUPEDDEE
Federal University of Santa Maria
Santa Maria, Brasil
vinicius-descovi-rodrigues@ufsm.br*

João Vitor Merlo
Graduate Program in Electrical Engineering (PPGEE)
Federal University of Santa Maria (UFSM)
Santa Maria, RS, Brazil
Email: joao.merlo@acad.ufsm.br

Abstract—This paper proposes a discrete-time Model Adaptive Controller (MRAC), applied for a Half-Bridge synchronous buck converter for charging batteries. The system uses an LCL filter that interfaces the power converter and the batteries. A virtual plant method is proposed for better transitory response. The model reference utilizes a reduced order, to simplify the project of the adaptive controller. Simulation results are presented for validation of the control project.

Index Terms—Adaptive Control, Virtual Plant, LCL filter,

I. INTRODUCTION

When direct current (DC bus) is the available power source, DC-DC converters are employed. Usually, this power comes from a converter, which is a circuit made up of rectifier diodes or a battery bank, and is intended to power a load that needs a variable DC voltage. Frequency modulation (FM) and pulse-width modulation (PWM), among other power semiconductor device switching techniques, are used to convert a fixed DC voltage source into an adjustable DC voltage source [1] [2]. The voltage reduction module or DC/DC BUCK converter is powered by a constant value and ripple-free continuous voltage source. With the MOSFET or IGBT switch, known as a power switch, the converter is capable of supplying the load with a fixed value of continuous voltage, without any polarity inversion, and with a voltage value lower than the input supply voltage of the circuit. Due to this electronic switching, the DC/DC buck converter is also referred to as a switched-mode power supply [1].

The deployment of adaptive control in DC-DC converters with LCL filters is a highly robust and effective method for handling the dynamic and variable behavior of these devices. The real-time nature of adaptive control ensures that the system remains stable and performs excellently even under stringent operating conditions. Consequently, this technology is particularly useful for critical installations, such as in energy storage systems, where efficiency and reliability are crucial factors [3], [4].

A common component of photovoltaic (PV) energy systems is the buck converter. The fluctuations in sun irradiation can cause a large range in the output of photovoltaic panels. Buck converters aid in output voltage regulation, supplying a steady power source for loads or battery storage. These

converters help maximize conversion efficiency, making them a dependable option for solar-powered systems that are both freestanding and grid-connected. Batteries are often utilized in renewable energy systems to store excess energy generated during peak production times for later use during low production periods. DC-DC converters, such as buck converters, are used to control the charging and discharging of batteries, ensuring that voltage levels are appropriate for safe and efficient operation. This application is critical for maintaining the balance and dependability of renewable energy systems, particularly in off-grid or hybrid settings [5], [6].

DC-DC converters are utilized in a variety of applications other than renewable energy systems, such as electric vehicles, distributed DC systems, electric traction and machine tools, and fuel cells. In electric vehicles, DC-DC converters are used to convert battery voltage to the levels required by various vehicle components. In distributed DC systems, these converters efficiently transfer electricity. They also regulate the power sent to motors and other machinery in electric traction and machine tools. In fuel cell systems, DC-DC converters transform the fuel cell's low voltage output to a higher voltage level suited for practical uses [5]. In remote locations with limited access to energy, independent solar pumping devices provide a viable water delivery solution. The buck converter decreases the voltage from PV arrays to a level that pumping systems can use efficiently, assuring a consistent and long-term water supply [6].

Considering the vital role that DC-DC converters play in these various applications, especially in renewable energy systems, it is imperative to create strong control strategies to guarantee the effective and dependable functioning of these converters. In order to maximize energy conversion, maintain the durability and safety of the componentry, and manage the dynamic and variable character of renewable energy sources, effective control techniques are required. This prepares the ground for investigating adaptive control strategies, which present a viable way to improve the stability and performance of DC-DC converters in a range of operating environments [7], [8].

Additionally, control strategies like adaptive control are critical to enhancing the efficiency and performance of buck

converters, which are vital components in the management of power and voltage in a variety of applications. In order to provide efficient power delivery and real-time stability, these control systems dynamically modify the operation of buck converters in response to changes in load circumstances and input voltages. Buck converters can maintain optimal output even in the face of input volatility by using adaptive controls, which greatly improves system efficiency and dependability. This focus on adaptive controls reflects the increasing demand for complex management techniques in power electronics and highlights their critical role in improving the performance and dependability of buck converters [9].

This paper proposes a discrete-time Model Adaptive Controller (MRAC), applied for a Half-Bridge synchronous buck converter for charging batteries. The system uses an LCL filter that interfaces the power converter and the batteries, the passive damping method with a series resistor is used for attenuation of the filter resonance peak. A virtual plant method is proposed for better transitory response. The model reference utilizes a reduced order, considering that the capacitor of the filter is zero, to simplify the project of the adaptive controller. Simulation results are presented for validation of the control project and the virtual plant effectiveness.

II. POWER CONVERTER AND PLANT

The topology used in this work is a Half Bridge converter, that functions like a synchronous Buck converter. As shown in Figure 1, the power converter is feed by a external DC Voltage Bus, and a LCL filter with a damping resistor is used for make the connection between the converter and the batteries, the project of the converter and damping resistor is proposed in [10], the Table I presents the values of the plant.

TABLE I
PARAMETERS OF POWER CONVERTER AND PLANT

Description	Variable	Value
Maximum battery voltage	V_{b_max}	16,8V
Minimum battery voltage	V_{b_min}	10V
Maximum current charge	$I_{b_max_Charge}$	2,6A
DC BUS Voltage	V_{cc}	24V
Converter side Inductor	L_1	60 μ H
Battery side Inductor	L_2	20 μ H
Filter Capacitor	C	86 μ F
Damping Resistor	R_d	0.5 Ω
Internal battery resistance	R_b	100m Ω
Switching Frequency	f_{sw}	50kHz

A. Modeling of LCL Filter

The LCL filter is commonly applied for grid-connected inverters, for attenuation of current harmonic components. This is a third-order filter, in comparison with a first-order filter, such as the L filter, that presents better attenuation of the harmonics provided by the switching frequency [11] [12]. The LCL filter has an attenuation of 60 dB/déc., while the L filter 20 dB/déc., despite the more complex topology, the LCL filter can be constructed with less volume of reactive components [13] [14].

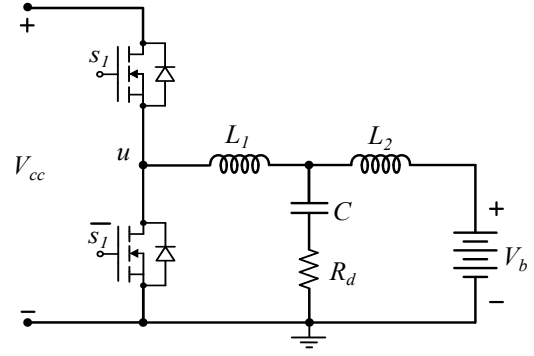


Fig. 1. Half Bridge Synchronous Buck converter, LCL filter and battery

However, the LCL filter has a peak of resonance that can make the closed-loop control difficult [15]. There are two main ways to mitigate the resonance problem: passive damping and active damping. The passive damping methods use a passive damping resistor, or a combination of more passive components [16], this is a good straightforward solution, but the project of the damping resistor is crucial for minimizing losses and provide the resonance peak attenuation [17]. The active damping methods include single or multi-loops methods, which use digital filters and feedback loops [18] [19], that require complex control techniques, and can elevate the computational cost.

In this application, the LCL filter is used for connect the power converter and batteries. The passive damping methodology is used by adding a damping resistor R_d in series with the capacitor C of the LCL filter. The Figure 2 presents the equivalent circuit, considering the converter apply a mean voltage based in the action of control u , the battery model is based on a zero-order internal resistance and a voltage source.

For the battery charge current control, was opted to control the L_2 current, witch is the battery current. The transfer function $G_1(s)$ relates the L_2 current with the voltage control action u , is given by the Equation 1, is considering the contribution of the battery resistance R_b , and for the voltage V_b is zero.

$$G_1(s) = \frac{i_{L_2}(s)}{u(s)} = \frac{\frac{R_d}{L_1 L_2} s + \frac{1}{L_1 L_2 C}}{s^3 + \frac{(L_1 R_d + L_1 R_b + L_2 R_d)}{L_1 L_2} s^2 + \frac{(L_1 + C R_d R_b + L_2)}{L_1 L_2 C} s + \frac{R_b}{L_1 L_2 C}} \quad (1)$$

B. Reduction of the Plant Order

With the aim of facilitating the adaptive control project, the plant was modeled as the Equation 2, where $G_0(s)$ é the modeled part of the plant and $\mu \Delta_\alpha(s)$ is a additive dynamic [20].

$$G_1(s) = G_0(s) + \mu \Delta_\alpha(s) \quad (2)$$

The part of order reduction $G_0(s)$ is modeled as first order transfer function, and is obtained considering that the

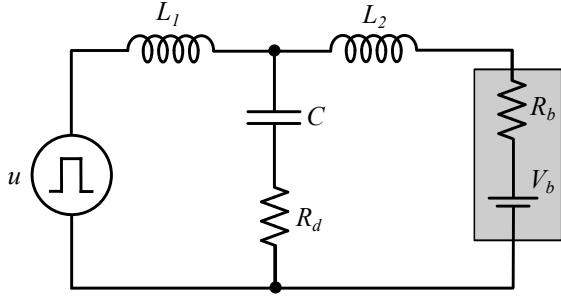


Fig. 2. Equivalent circuit of the converter, LCL filter and battery model

capacitance $C = 0$, the result is the transfer function of the Equation 3, which corresponds to a real pole of the plant. The pair of conjugated complex poles, is present in the additive non modeled dynamic $\mu\Delta_\alpha(s)$.

$$G_0(s) = \frac{\frac{1}{L_1+L_2}}{s + \frac{R_b}{L_1+L_2}} \quad (3)$$

C. The Virtual Plant Method

For the modeling of the plant considering the disturbance of the battery voltage, in Equation 4 is present the transfer function $G_2(s)$ that relates the L_2 current and the battery voltage V_b . The contribution of the power converter through the control action u is considered zero.

$$G_2(s) = \frac{i_{L_2}(s)}{V_b(s)} = \frac{\frac{1}{L_2}s^2 + \frac{R_d}{L_1L_2}s + \frac{1}{L_1L_2C}}{s^3 + \frac{(L_1R_d + L_1R_b + L_2R_d)}{L_1L_2}s^2 + \frac{(L_1 + CR_dR_b + L_2)}{L_1L_2C}s + \frac{R_b}{L_1L_2C}} \quad (4)$$

The sum of the transfer functions, as shown in Equation 5, represents the model of the plant, obeying the superposition law.

$$G(s) = G_1(s)u(s) + G_2(s)V_b(s) \quad (5)$$

This model of the plant, represented by the transfer function expressed in Equation 5, is a virtual model, that can be running in the DSP (Digital Signal Processor) before the connection with the real plant.

D. The Z transformation

For the discrete-time model of the plant, was used the Z transform and considering that the average voltage of the converter is kept constant between sampling events. This is modeled as a ZOH (Zero Order Hold). A delay of a sample related to digital implementation is also inserted, as shown in Equation 6.

$$G(z) = Z \left\{ e^{-sT_s} G(s) \right\} \frac{1}{z} \quad (6)$$

The Equations 7 and 8 represent the discrete model of the transfer function that relates the current i_{L_2} and the control

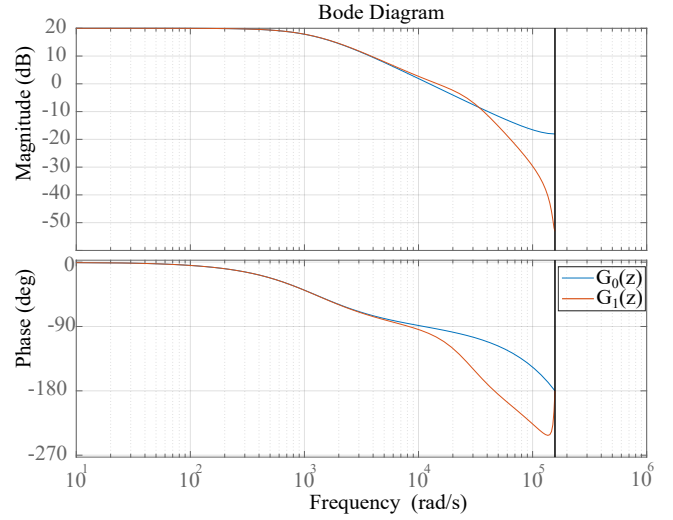


Fig. 3. Bode diagram of $G_0(z)$ and $G_1(z)$.

action u , with the values of the Table I with sample time $T_s = 20 \mu s$, $G_1(z)$ is the discrete nominal plant and $G_0(z)$ is the reduced model.

$$G_0(z) = \frac{i_{L_2}(z)}{u(z)} = \frac{0.2469}{z - 0.9753} \quad (7)$$

$$G_1(z) = \frac{i_{L_2}(z)}{u(z)} = \frac{0.0745z^2 + 0.02037z - 0.04218}{z^3 - 2.239z^2 + 1.709z - 0.4646} \quad (8)$$

As shown in Figure 3, the model of nominal plant $G_1(z)$ and the reduced order $G_0(z)$ are similar in low frequencies, and $G_1(z)$ have the passive damping by the resistor R_d that mitigate the peak of resonance of the filter.

The discrete model of the transfer function that relates the current i_{L_2} and the battery voltage V_b , is represented in the Equation 9.

$$G_2(z) = \frac{i_{L_2}(z)}{V_b(z)} = \frac{-0.7356z^2 + 1.302z - 0.6194}{z^3 - 2.239z^2 + 1.709z - 0.4646} \quad (9)$$

By applying the cross product in the Equations 8 and 9, it is possible to obtain the computable recurrence equation.

III. CONTROL METHODOLOGY

DC-DC converters are widely used in applications where efficient conversion of continuous voltage levels is necessary. The use of LCL filters in conjunction with these converters helps to reduce high-frequency harmonic components, thereby improving output quality and preventing electromagnetic interference. However, adding LCL filters introduces a series of problems, such as filter resonance, which can impair system stability. Thus, adaptive control is a good solution for this type of problem, as it can dynamically adapt to changes occurring in a system, maintaining the desired stability and performance. Adaptive control in LCL-filtered DC-DC converters refers to the real-time adaptation of controller parameters for robust

performance with changing operating conditions. This is especially required in applications where the nature of the load dynamically changes over time; for example, in an energy storage application, the energy source might be batteries [21].

Case studies and simulations have demonstrated that adaptive controllers are highly capable of improving the performance of DC-DC converters with LCL filters. Additionally, they have a strong ability to maintain system stability during transient operational and load variations. For instance, a simulation conducted in PSIM showed that adaptive controllers can dynamically adjust parameters to achieve a well-regulated converter output and minimal ripple current [21].

A. Adaptive Control

Model Reference Adaptive Control (MRAC) is based on the idea of designing an adaptive control system that makes the plant's output follow the output of a predefined reference model. Considering the plant G_p as first-order, this model describes the relationship between the input $U(s)$ and the output $Y(s)$ of the plant. To ensure that the plant exhibits the desired behavior, we define a first-order reference model, as presented in Equation 10 [22], in this application was chose reference model with a real pole in 1000 rad/s.

$$W_m(s) = \frac{Y_m(s)}{R(s)} = \frac{b_m}{s + a_m} \quad (10)$$

where $Y_m(s)$ is the desired output of the reference model in response to the input $R(s)$.

By applying the Laplace transform to the reference model, we obtain the differential equation in the time domain, as shown in Equation 11.

$$\dot{y}_m(t) = -a_m y_m(t) + b_m r(t) \quad (11)$$

The control equation is provided in Equation 12:

$$u(t) = \theta_1(t)r(t) - \theta_2(t)y(t) \quad (12)$$

where $\theta_1(t)$ and $\theta_2(t)$ are adaptive parameters that are dynamically adjusted to minimize the tracking error, as per Equation 13.

$$e(t) = y(t) - y_m(t) \quad (13)$$

The control action is expressed as:

$$u = \theta^T \omega \quad (14)$$

where $\omega = [\omega_1 \ \omega_2 \ y \ r \ V_b]^T$, and $\omega_1, \omega_2 \in \mathbb{R}^{n_p-1}$. The gain vector θ is given by $\theta = [\theta_1 \ \theta_2 \ \theta_y \ \theta_r \ \theta_{V_b}]^T$. The component V_b and θ_{V_b} are related to battery disturbance component.

The control gains θ can be generic expressed by the Equation 15.

$$\theta(t) = \theta(t-1) - \frac{T s \gamma \zeta(t-1)}{m^2(t-1)} \quad (15)$$

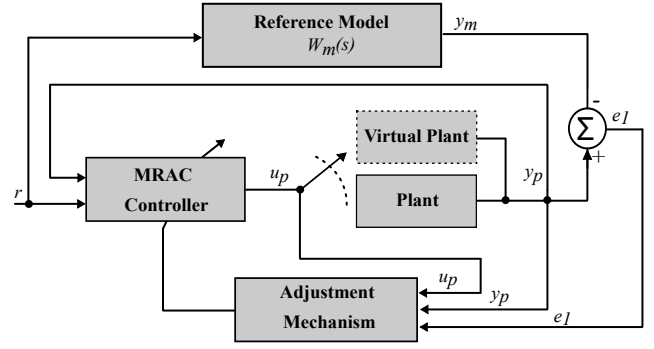


Fig. 4. Structure of the MRAC controller with the virtual plant method.

Where γ is a constant number for accelerate the adaptive gains, and m^2 is the normalizer. The updates of ω_1 and ω_2 are performed through the following equations:

$$\dot{\omega}_1 = F\omega_1 + qu \quad (16)$$

$$\dot{\omega}_2 = F\omega_2 + qy \quad (17)$$

The normalizer $m^2(t)$ is used to prevent large variations in the adaptive parameters and ensure system stability, as described in Equation 18.

$$m^2(t) = 1 + \zeta^T \zeta \quad (18)$$

where ζ is a vector containing relevant input and output signals.

The disturbance of the battery voltage is taken account in the control action by the vector gain θ_{V_b} . The Equation 19 is added to the control law.

$$u_{total}(t) = u(t) + \theta_{V_b} \quad (19)$$

The Figure 4 presents a simplified block diagram of the MRAC structure, considering the virtual plant.

IV. SIMULATION RESULTS

For the simulation results, was used the software PSIM. The circuit of the converter was assembled utilizing the specifications of component values as shown before. The MRAC controller and the virtual plant are executed in the C block functionality of PSIM, which emulates the code in DSP.

The results shown in Figure 5, present a sequence of steps in the current reference. Before 0.2 s, the system runs with the virtual plant (Y model), in 0.2 s the connection with the real plant is realized. The reference of current (Y_m) was 1 A for both instances, it is noted that the current has an overshoot when the real plant (i_{L2}) is connected, but adaptive gains settled minimize the effect for this condition of commutation.

In Figure 6, shows the starting transient where the virtual plant (Y model) is running, and the gains of control θ are adapted for the reference (Y_m), the adaptation gain γ utilized was 4000.

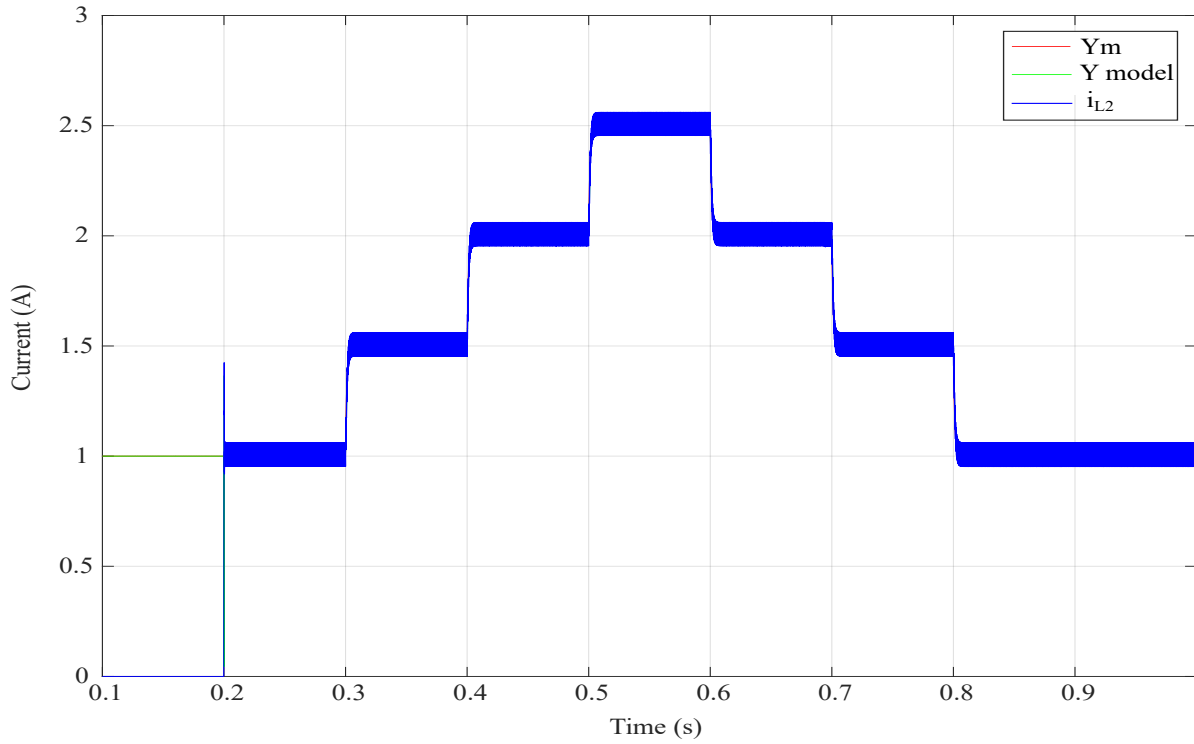


Fig. 5. Sequence of steps in current reference, virtual plant commutation.

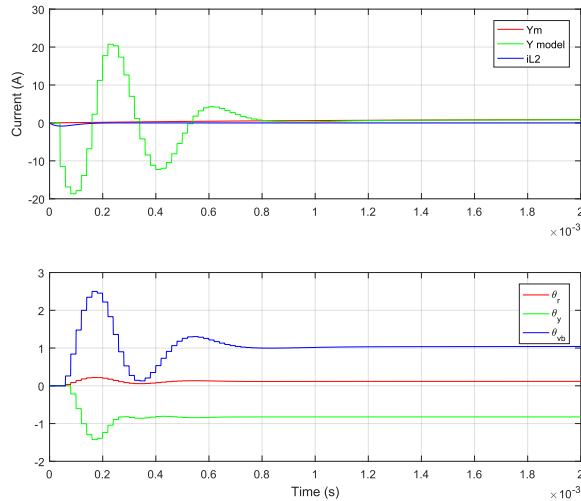


Fig. 6. Initial transitory with the virtual plant (Y model) running, and adaptive gains θ settling time.

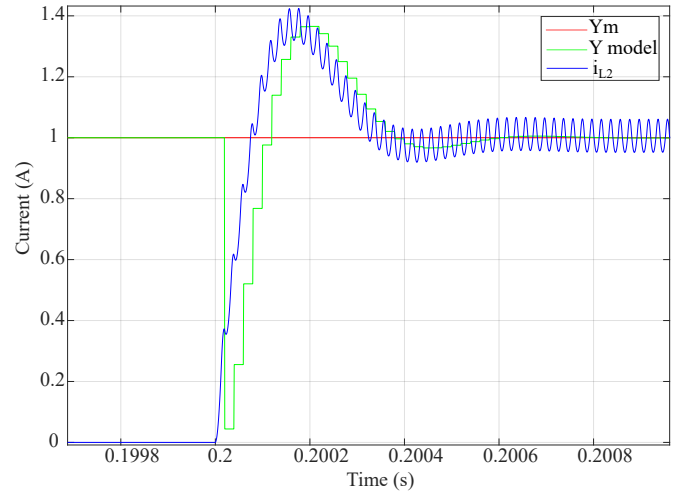


Fig. 7. Transitory between the virtual plant before $t = 0.2s$ and the connection with the real plant.

Figure 7 presents the moment of transition between the virtual plant (Y model) and the connection with the real plant (i_{L2}). It is possible to obtain a smoother transitory response since the θ gain vector is pre-tuned.

V. CONCLUSION

This paper has presented a discrete-time MRAC controller with a virtual plant method applied to a static converter with LCL filter for charging batteries. The principal contribution of this paper is to propose the virtual plant method for the use of

the MRAC controller for tracking DC references. Simulation results were presented for validation of the method.

ACKNOWLEDGMENT

REFERENCES

- [1] D. A. Arrabaça and S. P. Gimenez, *Conversores de Energia Elétrica CC/CC para Aplicações em Eletrônica de Potência - Conceitos, Metodologia de Análise e Simulação*. SRV Editora LTDA, 2013, e-book.
- [2] D. Hart, *Eletrônica de Potência: Análise e Projetos de Circuitos*. McGraw Hill Brasil, 2016.

- [3] A. Kumar, M. V. Aware, B. S. Umre, and M. A. Waghmare, "An adaptive multi-resonant current controller for a single-phase grid-tied converter with grid disturbance rejection capability," *IEEE Access*, vol. 10, pp. 28 053–28 065, 2022.
- [4] Y. Yin, J. Liu, W. Luo, L. Wu, S. Vazquez, J. I. Leon, and L. G. Franquelo, "Adaptive control for three-phase power converters with disturbance rejection performance," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 2, pp. 674–685, 2021.
- [5] N. B. Dawood, "Review of different dc to dc converters based for renewable energy applications," *International Research Journal of Engineering and Technology*, vol. 3, no. 3, pp. 46–50, 2016.
- [6] N. H. Baharudin, T. Mansur, F. A. Hamid, R. Ali, and M. I. Misrun, "Topologies of dc-dc converter in solar pv applications," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 8, no. 2, pp. 368–374, 2017.
- [7] M. Triggianese, H. Carbonnier, F. Tonicello, and M. R. Rogina, "Two-domains control of a buck converter," in *E3S Web of Conferences*, vol. 16. EDP Sciences, 2017, p. 14005.
- [8] H.-H. Chou, J.-Y. Chen, T.-H. Tseng, J.-Y. Yang, X. Yang, and S.-F. Wang, "A new control scheme for the buck converter," *Applied Sciences*, vol. 13, no. 3, 2023. [Online]. Available: <https://www.mdpi.com/2076-3417/13/3/1991>
- [9] S. Bagherwal, M. Badoni, S. Semwal, and S. Singh, "Design and development of standalone solar photovoltaic battery system with adaptive sliding mode controller," *Int. J. Renew. Energy Res*, vol. 10, no. 1, p. 8, 2020.
- [10] V. D. Rodrigues, A. O. Damasceno, and J. R. Tibola, "Modelagem e controle de conversor com filtro lcl para carga e descarga de baterias de Íons de lítio," in *Proceedings of the 15th Seminar on Power Electronics and Control SEPOC*, Universidade Federal de Santa Maria. Santa Maria: UFSM, 2023.
- [11] M. Liserre, F. Blaabjerg, and S. Hansen, "Design and control of an lcl-filter-based three-phase active rectifier," *IEEE Transactions on Industry Applications*, vol. 41, no. 5, pp. 1281–1291, 2005.
- [12] D. M. Lima, B. M. Lima, L. A. Maccari, and J. E. Normey-Rico, "Kalman filter observers with harmonic disturbance estimation applied to a grid-connected lcl filter," in *2018 13th IEEE International Conference on Industry Applications (INDUSCON)*, 2018, pp. 730–737.
- [13] M. Liserre, F. Blaabjerg, and R. Teodorescu, "Grid impedance estimation via excitation of lcl-filter resonance," *IEEE Transactions on Industry Applications*, vol. 43, no. 5, pp. 1401–1407, 2007.
- [14] M. Su, B. Cheng, Y. Sun, Z. Tang, B. Guo, Y. Yang, F. Blaabjerg, and H. Wang, "Single-sensor control of lcl-filtered grid-connected inverters," *IEEE Access*, vol. 7, pp. 38 481–38 494, 2019.
- [15] Y. Han, Z. Li, P. Yang, C. Wang, L. Xu, and J. M. Guerrero, "Analysis and design of improved weighted average current control strategy for lcl-type grid-connected inverters," *IEEE Transactions on Energy Conversion*, vol. 32, no. 3, pp. 941–952, 2017.
- [16] J.-Y. Lee, Y.-P. Cho, H.-S. Kim, and J.-H. Jung, "Design methodology of passive damped lcl filter using current controller for grid-connected three-phase voltage-source inverters," *Journal of Power Electronics*, vol. 18, pp. 1178–1189, 07 2018.
- [17] R. Peña-Alzola, M. Liserre, F. Blaabjerg, R. Sebastián, J. Dannehl, and F. W. Fuchs, "Analysis of the passive damping losses in lcl-filter-based grid converters," *IEEE Transactions on Power Electronics*, vol. 28, no. 6, pp. 2642–2646, 2013.
- [18] L. A. Serpa, S. Ponnaluri, P. M. Barbosa, and J. W. Kolar, "A modified direct power control strategy allowing the connection of three-phase inverters to the grid through lcl filters," *IEEE Transactions on Industry Applications*, vol. 43, no. 5, pp. 1388–1400, 2007.
- [19] W.-Y. Sung, H. M. Ahn, J.-H. Ahn, C.-Y. Oh, and B. K. Lee, "Sensorless active damping method of lcl-filter in grid-connected parallel inverters for battery energy storage systems," in *2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017 - ECCE Asia)*, 2017, pp. 1425–1429.
- [20] P. J. D. O. Evald, R. V. Tambara, and H. A. Gründling, "A direct discrete-time reduced order robust model reference adaptive control for grid-tied power converters with lcl filter," *Eletrônica de Potência*, vol. 25, no. 3, p. 361–372, Sep. 2020. [Online]. Available: <http://dx.doi.org/10.18618/REP.2020.3.0039>
- [21] M. H. Durgante, "Controle adaptativo de corrente em conversores conectados na rede elétrica numa estrutura multimalha," 2014.
- [22] R. Tambara, *Controle Adaptativo: Teoria e Aplicação*. Editora UFSM, 2024. [Online]. Available: <https://books.google.com.br/books?id=mkcAEQAAQBAJ>